

Omar Nashirudeen
42317AZ807

1. 1 car, 3 goats

Probability of winning if you stick

$$P(\text{win by sticking}) = \frac{1}{4} = 0.25$$

Probability of winning if you switch

$$P(\text{car not chosen initially}) = \frac{3}{4}$$

$$\text{so, } P(\text{win}) = \frac{3}{4} \times \frac{1}{2} = \frac{3}{8} = 0.375$$

Does switching help?

$$\text{stick} = 0.25$$

$$\text{switch} = 0.375$$

so switching helps.

Advantage

$$3\text{-door version: } \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

$$4\text{-door version: } \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

so the advantage is smaller

2. A: first component good

A': first defective

B: second good

B': second defective

(a) At least one component is good
 $A \cup B$

(b) One good and one defective
 $(A \cap B') \cup (A' \cap B)$

3. Given probabilities

$$P(A) = 0.8$$

$$P(B|A) = 0.85$$

$$P(B|A') = 0.75$$

(c) Probability second component is good

$$P(B) = P(B|A)P(A) + P(B|A')P(A')$$

$$P(B) = 0.85(0.8) + 0.75(0.2)$$

$$P(B) = 0.68 + 0.15 = 0.83$$

$$P(B) = 0.83$$

⑥ Probability at least one component is good

$$P(A \cap B^c) = P(A^c)P(B^c|A^c) \\ = 0.2(0.25) = 0.05$$

Thus $P(\text{at least one good}) = 1 - 0.05 = 0.95$

⑦ $P(A)$ at most one good

$$P(A \cap B) = 0.8 \times 0.85 = 0.68$$

$$P(A \cap B^c) = 0.8 \times 0.15 = 0.12$$

$$P(A^c \cap B) = 0.2 \times 0.75 = 0.15$$

$$P(A^c \cap B^c) = 0.2 \times 0.25 = 0.05$$

$$0.12 + 0.15 + 0.05 = 0.32$$

$$P(A \leq 1 \text{ good}) = \frac{P(A \cap B^c)}{0.32} \\ = \frac{0.12}{0.32} = 0.375$$

⑧ $P(A|B)$

Bayes rule: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

$$P(A|B) = \frac{0.85 \times 0.8}{0.83}$$

$$P(A|B) = \frac{0.68}{0.83} \approx 0.819$$

⑨ $P(A \cap B) = 0.68$

$$P(A)P(B) = 0.8 \times 0.83 = 0.664$$

$$0.68 \neq 0.664$$

Not independent

⑩ $P(A \cap B) = 0$

$$P(A \cap B) = 0.68$$

4. $P(E) = 0.008$

$$P(C) = 0.001$$

$$P(F|E) = 0.98$$

$$P(F|C) = 0.45$$

$$a) P(F) = P(F|E) P(E) + P(F|C) P(C)$$

$$P(F) = 0.98(0.008) + 0.45(0.001)$$

$$P(F) = 0.00829$$

$$b) P(E \cap F) = P(E) P(F|E)$$

$$= 0.008 \times 0.98$$

$$= 0.00784$$

$$c) P(\text{Engine-caused failure}) = 0.00784$$

5. This is a serial problem. The entire system works if all sub systems work. So,

Given:

Reliability of A (R_A) = 0.96

Reliability of B (R_B) = 0.95

Reliability of C (R_C) = 0.95

$$P_{\text{system}} = R_A \times R_B \times R_C$$

$$P_{\text{system}} = 0.96 \times 0.95 \times 0.95$$

$$P_{\text{system}} = 0.8664 \text{ or } (86.64\%)$$

$$6. \text{ Total} = 140 + 105 + 120 = 365$$

a) Number of acceptable transistors $A \geq 128$

Total transistors = 365

$$P(A \cap \text{Acceptable}) = \frac{128}{365} \approx 0.3507$$

$$b) P(\text{Acceptable} | C) = \frac{140}{170}$$

$$= 0.8235$$

c) Total marginal transistors
 $= 10(A) + 5(B) + 5(C) = 20$

Number of Marginal transistors from B = 5

$$P(B | \text{Marginal}) = \frac{5}{20} = \frac{1}{4} = 0.25$$